

Experiment-23

Aim

To perform image sharpening using the Unsharp Masking technique by subtracting a blurred version of the image from the original image.

Procedure

1. Select and load the input image.
2. Convert the image to grayscale.
3. Create a blurred version of the original image using a Gaussian filter.
4. Subtract the blurred image from the original image to obtain the unsharp mask.
5. Add the unsharp mask to the original image.
6. Normalize/clip the pixel values to the range 0–255.
7. Display the original, blurred, unsharp mask, and sharpened images.
8. Observe that edges and fine details are enhanced in the sharpened image.

Python Code

```
import cv2
img = cv2.imread("image.jpg")
if img is None:
    print("Error: Image not found")
    exit()
blurred = cv2.GaussianBlur(img, (5, 5), 0)
sharpened = cv2.addWeighted(img, 1.5, blurred, -0.5, 0)
cv2.imshow("Original Image", img)
cv2.imshow("Blurred Image", blurred)
cv2.imshow("Sharpened Image", sharpened)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Sample Input:



Output:



Results:

The image was successfully sharpened using the Unsharp Masking technique. The blurred image was subtracted from the original image to obtain the unsharp mask, which was then added to the original image. The resulting image showed enhanced edges and fine details.